Analysis part4

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Describing diagnostic tests

Diagnostic tests can be evaluated and described by the performance of distinguishing between people with and without a condition (usually compared to a gold standard).

Table: Diagnostic test outcome table

	Disease Present	**Disease Absent**
Test Positive	✅ TP (true positive)	❌ FP (false positive)
Test Negative	❌ FN (false negative)	✅ TN (true negative)

Sensitivity → TP / (TP + FN) = of those with disease, how many tested positive

Specificity → TN / (TN + FP) = of those without disease, how many tested negative

Libraries and interesting values

```r
# install.packages(c("epiR", "caret", "pROC", "DescTools"))
library(epiR)
library(caret)
library(pROC)
library(DescTools)
```

```r
mat <- matrix(c(72, 8, 20, 100), nrow = 2, byrow = TRUE)
dimnames(mat) <- list(Disease = c("Yes", "No"), Test = c("Positive",
mat
```

```
##        Test
## Disease Positive Negative
##     Yes       72        8
##     No        20      100
```

Run the test

```
epi.tests(mat)
```

```
##           Outcome +    Outcome -      Total
## Test +           72            8         80
## Test -           20          100        120
## Total            92          108        200
##
## Point estimates and 95% CIs:
## ---------------------------------------------------------------
## Apparent prevalence *                  0.40 (0.33, 0.47)
## True prevalence *                      0.46 (0.39, 0.53)
## Sensitivity *                          0.78 (0.68, 0.86)
## Specificity *                          0.93 (0.86, 0.97)
## Positive predictive value *            0.90 (0.81, 0.96)
## Negative predictive value *            0.83 (0.75, 0.90)
## Positive likelihood ratio              10.57 (5.38, 20.76)
## Negative likelihood ratio              0.23 (0.16, 0.35)
## False T+ proportion for true D- *      0.07 (0.03, 0.14)
## False T- proportion for true D+ *      0.22 (0.14, 0.32)
## False T+ proportion for T+ *           0.10 (0.04, 0.19)
## False T- proportion for T- *           0.17 (0.10, 0.25)
## Correctly classified proportion *      0.86 (0.80, 0.90)
## ---------------------------------------------------------------
## * Exact CIs
```

Manually

```r
sensitivity <- 0.85
specificity <- 0.80

LR_pos <- sensitivity / (1 - specificity)
LR_neg <- (1 - sensitivity) / specificity
LR_pos
```

```
## [1] 4.25
```

```r
LR_neg
```

```
## [1] 0.1875
```

ROC curve and AUC

Receiver Operating Characteristic Curve

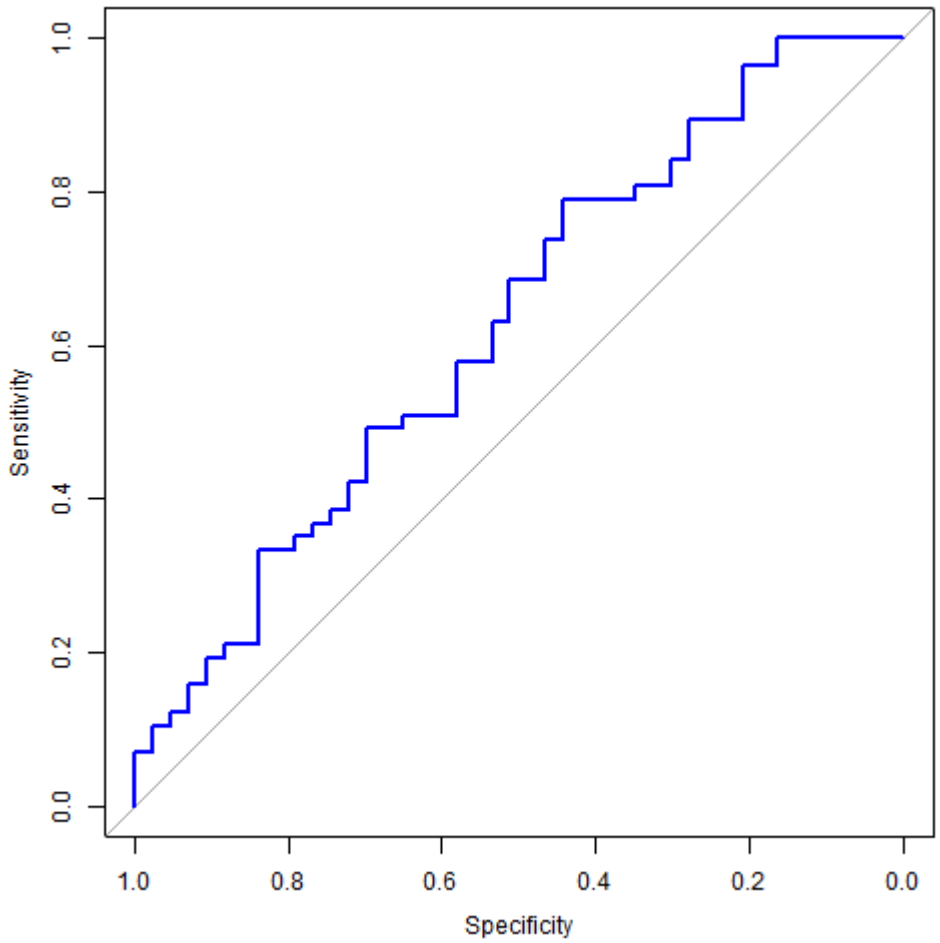
The ROC curve is a plot used to evaluate the performance of a binary classification test (e.g., disease vs. no disease).

It shows the trade-off between:

- Sensitivity (True Positive Rate) — on the y-axis
- 1 – Specificity (False Positive Rate) — on the x-axis

```
set.seed(123)
actual <- factor(sample(c("disease", "no_disease"), 100, replace = TR
score <- rnorm(100) + ifelse(actual == "disease", 1, 0.5)  # biomarke
```

```r
roc_obj <- roc(actual, score, levels = c("no_disease", "disease"))
plot(roc_obj, col = "blue")
```



```r
auc(roc_obj)
```

```
## Area under the curve: 0.6255
```

- An AUC of 0.62 means that the test has low but better-than-random discriminatory ability.
- The test has a 62% chance of correctly distinguishing a randomly chosen patient with the disease from one without the disease.

Comparing tests

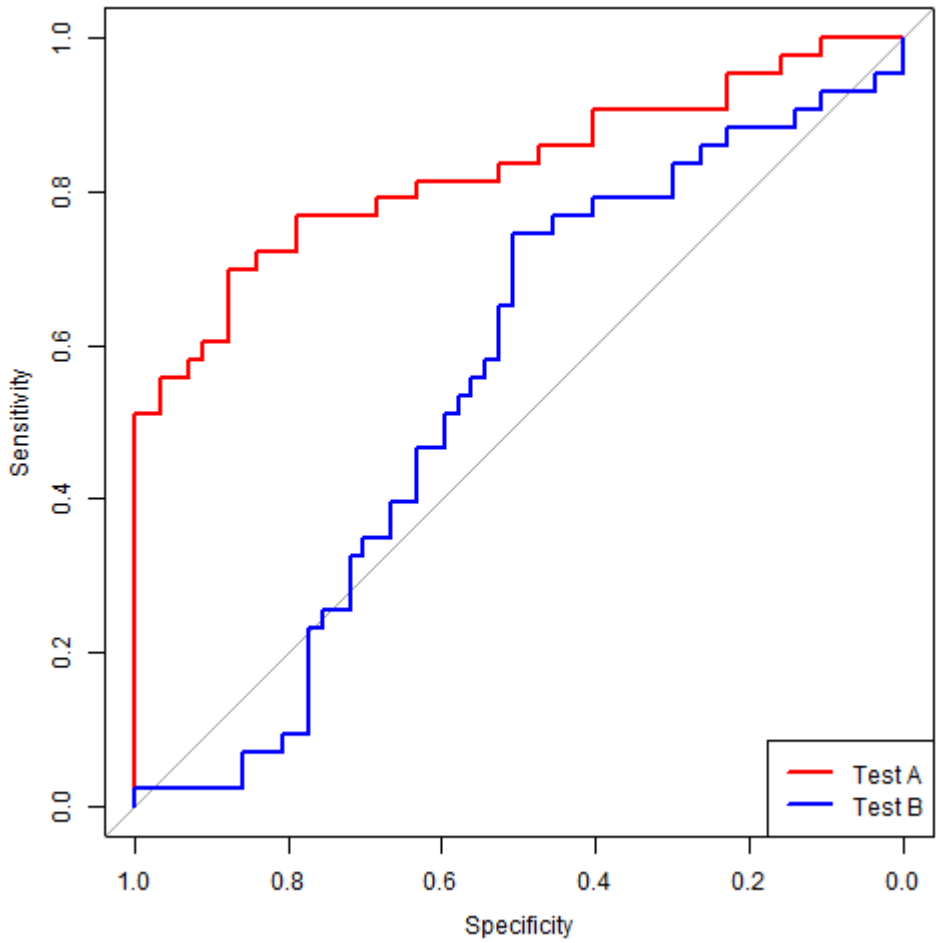
```r
# Simulate two test scores
testA <- rnorm(100) + ifelse(actual == "disease", 1.5, 0.5)
testB <- rnorm(100) + ifelse(actual == "disease", 1.0, 1.0)

rocA <- roc(actual, testA)
```

```
## Setting levels: control = disease, case = no_disease

## Setting direction: controls > cases
```

```r
rocB <- roc(actual, testB)
```

```
## Setting levels: control = disease, case = no_disease

## Setting direction: controls < cases
```

```r
plot(rocA, col = "red")
plot(rocB, add = TRUE, col = "blue")
legend("bottomright", legend = c("Test A", "Test B"), col = c("red",
```



```
auc(rocA)
```

```
## Area under the curve: 0.8266
```

```
auc(rocB)
```

```
## Area under the curve: 0.5516
```

AUC Value Interpretation

- 0.5: No discrimination (random)
- 0.6 – 0.7: Poor discrimination
- 0.7 – 0.8: Acceptable
- 0.8 – 0.9: Excellent
- above 0.9: Outstanding